



**DHANALAKSHMI SRINIVASAN
UNIVERSITY**

Samayapuram, Tiruchirappalli, Tamil Nadu - 621 112

SCHOOL OF ENGINEERING AND TECHNOLOGY



DEPARTMENT OF AI & DS

REVIEW-2 FORM

Academic Year: 2024 – 2025

EVEN Semester

Name (s) 1. Shaik Rakeeb Sha Valli 2. M Sri Anjaneyulu 3. B Shiridi Karthikeya	Register Number (s) 1.11523050665 2.11523050646 3.11523050633	Subject Code 21CSEMP
Year / Semester / Section: III/VI/B5		Team No.:5

Domain Name : Data analytics and Mechine learning

Title Name: **Real-Time AI-Powered Sales Analytics and Smart POS System with AI Voice Assistant**

List of Comments & Implementation Progress Report:

1. Module Description

- **Smart POS Cashier Terminal:** Built a high-performance billing interface featuring a live search catalog, barcoded barcode scans, active cashier shifts tracker, invoice receipt generator, and dynamic client discounts (Bronze, Silver, Gold, Platinum tiers).
- **Visual Admin Management Portal:** Developed a visual business intelligence dashboard rendering real-time metrics charts (Total Revenue, Transactions, Net Profit, Average Order Value), visual inventory status tables with quick reorder triggers, active shifts table, and live transaction audit log.
- **E-Commerce Customer Storefront:** Created a responsive luxury storefront featuring fluid cyberpunk styling, interactive storefront grids, persistent shopping carts, custom glassmorphic portals, and seamless return links to the main application.
- **AI-Powered Voice Assistant:** Integrated a natural-language semantic billing module using Groq Cloud LLM APIs to perform voice checkout, inventory details lookup, cashier shifts analysis, and VIP customer spent-amount queries.
- **CSV Bulk Inventory Importer:** Created a robust file-based CSV parser allowing cashier/admin to batch upload products catalog and update store state with interactive validation previews.

2. Coding

- **Frameworks & Libraries:** Developed entirely in React.js 18 and Vite for super-fast hot module

replacement (HMR), employing Zustand for highly responsive client-side state synchronization across POS and admin views.

- **Design System & UI:** Handcrafted with modern HTML5 semantic elements and Vanilla CSS, implementing glassmorphic cards, linear gradients, HSL colors, responsive grids, custom scrollbars, and modern typography (Outfit/Inter).
- **Database & Persistence:** Integrated Firebase / Firestore SDK for real-time transactions logging and authentication, supplemented by localStorage for shopping cart configuration, sidebar pinned state, and active cashier session persistence.
- **Advanced Canvas Animations:** Configured custom interactive mouse physics followers (lerp-based lag dots), click particle ripple bursts, and an ambient canvas node-connection networking background.

3. Implementation

- **Robust Architecture:** Implemented modular React architecture separating voice assistant context refs, data management components, storefront wrappers, and POS systems for easy maintenance and zero state-closure memory leaks.
- **Production Multi-Page Bundling:** Configured advanced multi-page rollup options in vite.config.js to compile index.html and customer-website.html concurrently into high-efficiency static assets with zero errors.
- **Asynchronous LLM Processing:** Integrated a dual-model LLM fallback processing pipeline (llama-3.3-70b-versatile and llama3-8b-8192) to guarantee natural language assistant uptime under rate limits.
- **GitHub Deployment:** Successfully committed and pushed all clean source code, assets, packages, and completed builds to GitHub: <https://github.com/Shaik-Rakeeb-Shavalli/Supermart.git>.

Name of the guide with signature	Mrs.Sheeba
Approved / Not approved	
Remarks, if any	

Project Co-ordinator

HOD